Like, Comment, Get Scammed: Characterizing Comment Scams on Media Platforms

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***Abstract*—The research uncovers elaborate scamming strategies used by fraudsters, presenting thorough examination of comment scam networks. The authors of this study created a system to gather millions of YouTube comments and used different filters to identify scam campaigns directing users to contact attackers. Besides this work, multiple other studies have examined the same problem, but from more limited, yet complementary viewpoints. The investigation I conducted focuses on assessing the results from research about spam detection and filtering techniques.**

***Index Terms*—social media security, comment scams, scam detection, user protection, platform moderation**

I. Context

YouTube became the largest video sharing platform since its establishment in 2005 because over 400 hours of new content are uploaded daily while 4.5 million videos get watched every minute (Statistics and Facts n.d). The widespread availability of YouTube videos has led to an increase in spam comments that typically serve to promote other channels or videos within the comment sections of popular content. However, YouTube's spam filter remains ineffective despite its presence on the platform.

II. Solution Description

For this project, the researchers designed a specialized infrastructure to longitudinally capture YouTube comments over 6 months.

A. Research Methodology and Infrastructure

1) **Crawl Module:** It automatically retrieves comments of interest using the Google YouTube Data API. The tool does not snap and collect static snapshots, but rather runs hourly to capture comments that appear and disappear in an ecosystem simulation.
2) **Storage Module:** It maintains comments as well as metadata, and each comment's content is archived alongside its origin timestamp, user channel ID, and corresponding account profile image. In order to store comments from multiple channels separately, a custom database was designed.
3) **Analysis Module:** It utilizes three specialized filters to detect scam-related activity.
   a) **Text-based Filter:** Finding phone numbers, mentions of Telegram and WhatsApp, and especially the use of visually similar symbols (VSS) by iteratively refining a list of suspicious symbols using the so-called 'snowball approach'.
   b) **Image-based Filter:** Identifying impersonation via profile image similes by hashing to check if the user's profile image matches or closely resembles the real channel owner's image.
   c) **Time-Based Filter:** Recognizing patterns of scripted conversations and coordinated commenting from dozens of accounts in blinding speeds in seconds.

B. Key Findings

This study raises several concerns about scam comments on YouTube:

1) **Poor Detection.** Current spam detection systems overlook visually disguised and impersonator accounts posing as the channel's official owner.
2) **Adaptive Strategies.** Scam methods are evolving; for instance, the use of bot accounts that post scripted replies in a "conversational" style, often within 15-second intervals to appear authentic.
3) **Financial Damage.**
   - Victims suffer direct losses from cryptocurrency or digital payment scams after being persuaded to send money to fake "financial gurus" or to pay "shipping fees" for nonexistent prizes.
   - Blockchain analysis confirmed $1M–$2M in stolen funds from just 50 scammers asking for cryptocurrency payments.
4) **Scam Prevalence**
   - Approximately 2.34% (i.e., 206,306 comments) were flagged as scams, which means over ten thousand unique accounts posted an average of 1,140 scam comments daily.

- Although finance channels were the most targeted, cooking and political channels were also significantly affected by spam.

5) **Scam Types**
   - **Investment Scams:** Commonly linked to cryptocurrencies or trading, these scams promise high returns.
   - **Fake Prize Scams:** Users are informed they've "won" an expensive prize, but must pay "fees" or "shipping" to claim it.
6) **Scammer Lifespan**
   - Roughly 31.42% of identified scammer accounts were deactivated within six months, but the majority persisted longer by recycling or rotating accounts.
   - About 59.87% of scam comments were eventually removed—by the channel owner, the authors themselves, or YouTube.

III. Related Work

This section reviews previous research in the domain of online scams, especially those that exploit social media platforms.

A. Crypto Fraud vs. Comment-Based Deception

Bartoletti and his colleagues [5] study cryptocurrency scams (Ponzi, ICO fraud) and discuss the general characteristics of the fraudulent projects as well as partial financial damages incurred. Li et al. also research scams in crypto, but more suitable to a platform is they link the scams to YouTube comment threads, thus uncovering how fraudsters employ such scams to lure victims in transferring money. The quick association between social-engineering and crypto outflows in Li et al.'s compelling victim-impact study is unknown.

B. From Generic YouTube Spam to Sophisticated Scam Networks

Alberto et al. [1] devised a machine-learning-based filter (Naïve Bayes, SVM) for YouTube comment spam, focusing on textual features and accuracy metrics. While they share Li et al.'s mission—YouTube spam detection—the 2015 paper targets generic spam (ads, repetitive messages). By contrast, Li et al. delve into scam networks, phone/Telegram obfuscation, and real user losses.

C. Early Data-Mining Approaches

Chowdhury et al. [4] present an early data-mining system for YouTube spam detection, applying classification trees and clustering based on text features and limited account metadata (e.g., age, ratio of replies). Both lines of work rely on large-scale data from YouTube. Chowdhury's model, however, remains a short-term detection framework. Li et al., by contrast, compile a longitudinal dataset, integrate multi-filter strategies, and detail financial theft. The Li et al. approach thus surpasses basic spam filtering with richer scammer-behavior insights.

D. Cross-Platform Abuse

Costin et al. [6] analyze phone numbers in cybercrime, illustrating how attackers repeatedly reuse contact lines across malicious campaigns. This resembles Li et al.'s observation that scammers revolve around phone/Telegram details. Yet Costin et al. emphasize cross-platform phone usage rather than YouTube comment infiltration. Li et al. refine phone-based detection by linking contact info to rapid-fire conversation scripts and impersonation in a public forum. Hence, Li et al. meaningfully extends the phone-centric lens—merging textual, graphical, and timing filters in YouTube comment environments for a more holistic approach.

E. Video-Level Monetization vs. Comment Fraud

Tripathi et al. [7] analyze monetizing scam videos on YouTube, focusing on suspicious creators uploading get-rich-quick content for ad revenue or affiliate links. Both Tripathi and Li et al. highlight YouTube-based social-engineering, but the former emphasizes video-level deception and metadata, while Li et al. concentrate on comment-level scams, harnessing textual, temporal, and image-based cues to detect orchestrated campaigns and measure stolen crypto.

F. Online Spam Detection

Within YouTube-specific research, most prior work classified comments into binary classes—spam or ham—without addressing the complexity of coordinated scams [1], [2]. Oh et al. [3] applied ensemble machine learning models to detect spam comments on YouTube with impressive accuracy, but their scope was limited to detection rather than examining scammer behaviors, tactics, or financial impact. Unlike this approach, the current study provides a holistic characterization of comment scams including their tactics, campaign structures, and economic impact.

G. Key Differences and Contributions

Compared to existing research in social-media-based scams, the paper by Li et al. offers a holistic perspective that includes:

1) Longitudinal monitoring of major YouTube channels and all comment activities.
2) A multi-filter detection mechanism fusing textual, visual, and temporal features for high-precision scam identification.
3) Direct evidence of scammer tactics and profits through real conversations and blockchain transaction analysis—lending concrete figures to the scope of the scam.

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